Regression Interview Questions

Name: Satish G40 AI&ML

1. What is regression in machine learning, and when is it used?

Regression is a supervised machine learning technique used to predict continuous numerical values based on input data. It identifies the relationship between independent variables (features) and a dependent variable (target). Regression is commonly used for predicting values such as house prices, salaries, sales, temperature, and stock prices.

2. What is the difference between simple linear regression and multiple linear regression?

Simple Linear Regression uses one independent variable to predict a dependent variable, while Multiple Linear Regression uses two or more independent variables for prediction. Multiple regression generally provides more accurate predictions when several factors influence the target variable.

3. What assumptions are made by linear regression?

Linear regression assumes:

- There is a linear relationship between independent and dependent variables.
- The observations are independent of each other.
- The variance of errors is constant (homoscedasticity).
- The errors are normally distributed.
- There is little or no multicollinearity among independent variables.
- There are no significant outliers affecting the model.

4. What is the difference between a dependent variable and an independent variable?

An independent variable is the input feature used to make predictions. A dependent variable is the output or target variable whose value is predicted based on the independent variables.

5. What do the slope and intercept represent in a linear regression model?

The slope represents the amount by which the dependent variable changes for every one-unit increase in the independent variable. The intercept represents the predicted value of the dependent variable when the independent variable is zero.

6. What is the purpose of the cost function in regression?

The cost function measures the difference between the actual values and the predicted values. Its purpose is to evaluate the performance of the regression model and guide the optimization algorithm to find the best-fitting model by minimizing prediction errors.

7. What is Mean Squared Error (MSE), and why is it commonly used for regression?

Mean Squared Error (MSE) is the average of the squared differences between actual and predicted values. It is commonly used because it penalizes larger errors more heavily, making the model more sensitive to significant prediction mistakes. A lower MSE indicates better model performance.

8. What is the difference between Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE)?

- **MAE (Mean Absolute Error):** Measures the average absolute difference between actual and predicted values. It is simple to interpret and less affected by outliers.
- **MSE (Mean Squared Error):** Measures the average squared difference between actual and predicted values. It gives more weight to larger errors.
- **RMSE (Root Mean Squared Error):** Is the square root of MSE and is expressed in the same units as the target variable, making it easier to interpret.

9. What is the R-squared score, and how should it be interpreted?

R-squared (R^2) measures how well the regression model explains the variation in the dependent variable. Its value ranges from 0 to 1. A higher R^2 value indicates that the model explains a larger portion of the data's variability, while a lower value indicates a weaker fit.

10. What is adjusted R-squared, and why is it useful?

Adjusted R-squared is a modified version of R-squared that considers the number of independent variables in the model. It helps determine whether adding new features actually improves the model or simply increases complexity without providing useful information.

11. What is overfitting in a regression model?

Overfitting occurs when a regression model learns not only the underlying patterns in the training data but also the noise. As a result, the model performs very well on training data but poorly on unseen or test data because it does not generalize well.

12. What is underfitting in a regression model?

Underfitting occurs when the regression model is too simple to capture the underlying patterns in the data. It results in poor performance on both training and testing data because the model fails to learn the relationship between the variables.

13. What is multicollinearity, and how does it affect regression?

Multicollinearity occurs when two or more independent variables are highly correlated with each other. It makes it difficult to determine the individual effect of each variable, leading to unstable coefficient estimates and reducing the interpretability of the regression model.

14. How can multicollinearity be detected?

Multicollinearity can be detected using:

- Correlation Matrix
- Variance Inflation Factor (VIF)
- Heatmaps
- Pair plots

A high VIF value or a strong correlation between independent variables indicates multicollinearity.

15. What is regularization, and why is it used?

Regularization is a technique used to prevent overfitting by adding a penalty term to the regression model. It reduces the complexity of the model by controlling the size of the regression coefficients, thereby improving the model's ability to generalize to new data.

16. What is the difference between Ridge, Lasso, and Elastic Net regression?

- **Ridge Regression:** Uses L2 regularization and reduces the size of coefficients without making them zero.
- **Lasso Regression:** Uses L1 regularization and can reduce some coefficients to exactly zero, effectively performing feature selection.
- **Elastic Net Regression:** Combines both L1 and L2 regularization, providing the benefits of Ridge and Lasso together.

17. How does Lasso regression perform feature selection?

Lasso regression performs feature selection by shrinking the coefficients of less important features to zero. Features with zero coefficients are effectively removed from the model, resulting in a simpler and more interpretable model.

18. What is polynomial regression?

Polynomial regression is an extension of linear regression that models nonlinear relationships by adding polynomial terms of the independent variables. It is useful when the relationship between the variables is curved rather than linear.

19. How do outliers affect a regression model?

Outliers can significantly affect a regression model by pulling the regression line toward them, resulting in inaccurate predictions and higher error values. They may reduce the overall performance of the model if not handled properly.

20. How would you evaluate the performance of a regression model?

The performance of a regression model can be evaluated using metrics such as:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R-squared (R^2)
- Adjusted R-squared